Culinary AI: Personalized Recipe Recommendations Powered by RAG

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***Abstract*—The increasing growth of personalized services in the culinary industry has created a need for advanced recipe recommendation systems. This study highlights the design and development of an intelligent recipe recommendation system that uses a Retrieval Augmented Generation approach that integrates information retrieval and generative capabilities. The traditional systems often fail to cater to the needs of the user when it comes to suggesting recipes with the available ingredients, our system addresses these issues by utilizing robust Natural Language Processing and Machine Learning techniques to enhance user experience. Our system is developed using a large dataset comprising over 490,000 recipes sourced from Kaggle. The system mainly uses ingredient-based vector embedding using Word2Vec, content-based filtering for personalization, and model fine-tuning to align recommendations with user preferences. For evaluation, we have collected user feedback via surveys and used those ratings on a scale of 10 to calculate the accuracy of the suggestions. By incorporating the RAG, our project aims to push the boundaries of the recommendation systems integrating it with the latest generative models and retrieval systems which can provide context-aware recipe suggestions and an efficient solution for the users looking new to cooking or seeking healthy meal options.**

***Index Terms*—Vectorization, Indexing, Retrieval, LLM, Generation.**

I. Introduction

Since the growth of digital platforms, people have become much more in need of having a personalized recipe list. People often search for well-developed recipes that meet their needs in terms of what they can consume or create. Still, people find it difficult to get a wonderful recipe though there are large number of recipes available in the database. By devising an advanced recipe recommendation system based on the Retrieval-Augmented Generation (RAG) architecture our project attempts to bridge the gap. To find related recipes and to understand the fundamental ideas behind them, the suggested approach combines content-based filtering, FAISS indexing and a generative model process. Datasets, a repository of Kaggle's that indicates basic structure of a recipe which includes names, necessary ingredients and instructional steps is a crucial part of this system.

The data will thus be helpful in improving the efficiency and usability of a recommendation system. The final product is made more engaging by including interactive and modern ideas like Natural Language processing (NLP), vectorization and retrieval techniques.

II. Problem Statement

Users often find it difficult to search for specific recipes from huge recipe databases. Unstructured data typically consists of inconsistent ingredient formats, undesired recommendations and lack of user instructions personalized for the individual are some of the difficulties. In addition to that, the majority of available recipe recommendation systems don't go deeply into the details that respond to user's query or the kinds of ingredients that are appropriate for the individual. Such a nature makes the process of finding new recipes extremely challenging and lowers customer satisfaction.

In order to address these issues, we are working on this project to create an effective recipe recommendation system. The dataset is processed and arranged to improve the database's validity and robustness. FAISS is then implemented for fast similarity searches and produces a model that offers precise explanations. With this, people who use our system will get detailed step-by-step instructions for relevant recipes, that help users to prepare a recipe without any mistakes.

III. Review of Literature

The paper by Maheshwari and Chourey (2019) presented an in-depth usage of machine learning-based recipe recommendation systems, which focused on ingredient-based recipe search [2]. The paper covered two models in which TF-IDF and Word2Vec were implemented. The main approach for the data collection involved web scraping the data to extract recipes and ingredient data from a single source. This paper provided an extended scope of the recipe recommendations system by utilizing the RAG models as it helps to search beyond a single website.

Wang et al. (2024) implemented RAG techniques to improve the performance of the LLMs [3]. The paper also delved into various other RAG components such as repacking, query classification, and summarization. The paper also proposed various strategies for enhance the functioning and the performance by the usage of a multimodal RAG.

The paper by Deldjoo et al. (2024) provided a comprehensive review of modern recommendation systems by using models such as GANs, LLMs and diffusion models [1]. The study traced back to the evolution of recommendation systems from a traditional user approach to an advanced framework.

The paper by Bai, X et al.(2022) said that in the domain of personalized recommendation systems, efficiently capturing user preferences and item attributes is critical for giving an accurate and satisfying recommendation [4]. This paper presents functionalities using LLMs to generate detailed textual descriptions. These authors proposed a dual approach that used a supervised fine-tuning combined with RAG to adapt to the users' needs and used LLMs knowledge base to enhance the item description and improve the system's overall performance.

According to the article by Shah, S et al.(2024) acknowledge the challenges an individual faces in meal planning caused by inventory invisibility and decision fatigue [5]. The authors proposed an AI-based solution that uses recognition of ingredients to give a personalized recipe recommendation based on the available items in the pantry, they also planned to simplify meal preparation, the system aims to reduce food wastage and promote healthier eating habits.

IV. Objectives of the study

- Using effective retrieval components (FAISS) and a variety of possible generative AI models, the main objective is to devise a more sophisticated recipe recommendation system that uses the Retrieval-Augmented Generation (RAG) framework.
- Our second objective is to enhance the user experience by presenting users with recipes that are appropriate to their needs by taking the ingredients as input that user has.
- To ensure structured recipe preparation with the focus on precision and clarity guaranteeing that the final product satisfies useful and user-friendly requirements regarding culinary activities.

V. Data Collection

A. Data Source

The dataset that we used for this project is gathered from Kaggle which is well known for data sharing and machine learning contests. Along with recipe names, there are also other things like how many ingredients are needed, preparation steps to make a recipe are available in the dataset. The selected dataset met all the requirements to build our recipe recommendation system.

B. Data Description

The dataset consists of 10 attributes. Each attribute represents a specific type.

1) **Recipe id:** It is for unique recipe identification.
2) **Recipe name:** It gives the information about the name of the recipe.
3) **Description:** This attribute gives the summary of each recipe.
4) **Ingredients:** It contains all the ingredients required to prepare a recipe.
5) **Ingredients raw str:** It gives information on the proportion required for each ingredient.
6) **Servings Size:** This gives information about how much protein you'll be getting with the recipe.
7) **Servings:** It refers to the number of individual servings that a recipe is designed.
8) **Steps:** Provides step-by-step instructions to prepare a recipe.
9) **tags:** It captures relevant labels.
10) **search terms:** It reflect user-driven queries that enhance search and context-based retrieval.

C. Dataset Link

Below is the kaggle website which we have used to obtain our data set.
https://www.kaggle.com/datasets/shuyangli94/foodcom-recipes-with-search-terms-and-tags

VI. Exploratory data analysis (EDA) and Hypotheses for the Study

A. EDA:

- **Data upload and Inspection:** For analysis, we have imported the dataset into jupyter notebook. Thorough inspections have been conducted on the dataset to identify any missing values, duplicate entries, and data inconsistencies.
- **Handling missing values and Duplicate values:** A total of 2 percent of the data was identified as either missing or duplicated across all the columns. So, to ensure data quality and consistency, these values have been removed.
- **Data Cleaning and Preprocessing:** We have also removed some unwanted columns like recipe id, description, ingredients raw str, servings, tags, and search terms as we dealt with only the ingredients column primarily.

In order to perform vectorization, we have removed unwanted characters like special symbols, white spaces to standardize the ingredient descriptions.

B. Hypotheses for the Study

The hypothesis we want to incorporate is the relevance of recipe recommendations which is the main idea of this project and is an important factor in the success of any recommendation system. In this paper we hypothesize that the Retrieval Augmented Generation model improves the relevance of the recipes suggested based on the inputs provided by the user. Specifically, the null hypothesis which states that the recipe recommendations generated by the RAG model are not relevant to the user's inputs, while the alternative hypothesis which suggest that the recommendations are highly relevant and suits the preference of the user. To evaluate this hypothesis, we will be collecting feedback through user surveys namely users rating the recipe suggested on the provide inputs on a scale of 10, scoring according to the perceived relevance of the recommended recipes.

VII. Methodology

A. Data Preprocessing and Vectorization:

Before modeling, we took care of cleaning the data

- To ensure uniformity and accuracy the dataset was preprocessed to handle missing and duplicate values. In this process we have removed 2 percent of the data. Special characters were removed from the ingredients column.
- In the next step, we have transformed the ingredients column into numerical representations using word2vec embeddings [7]. So, the textual data is converted into dense vectors that capture semantic relationships between the ingredients to retrieve the recipes accurately.

B. Model Architecture

1) Retrieval using FAISS:: For efficient vector-based retrieval we have used Facebook AI similarity Search (FAISS) that recommends recipes to the user.

- With the help of FAISS index, embeddings of recipes can be stored and retrieved.
- b) Based on cosine similarity [8], the system shows the top k most relevant recipes when a user inputs the ingredients list.

2) Generation using ChatGPT:: To provide users with a step-by-step guide on how to make the recommended recipes, we have used ChatGPT 3.5 Turbo model which is a popular LLM model [10].

- To generate natural language instructions for each suggested recipe, this model was integrated into the system.
- User experience will improve as ChatGPT will give clear and detailed instructions.

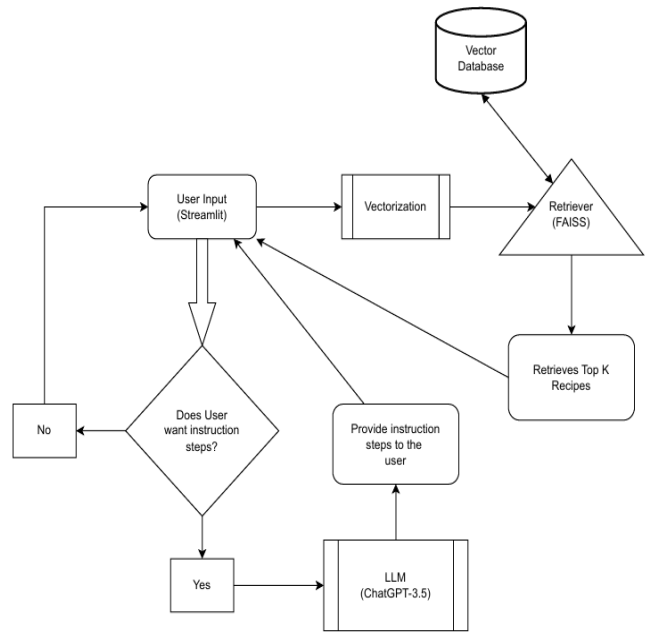

Fig. 1. Architecture Diagram

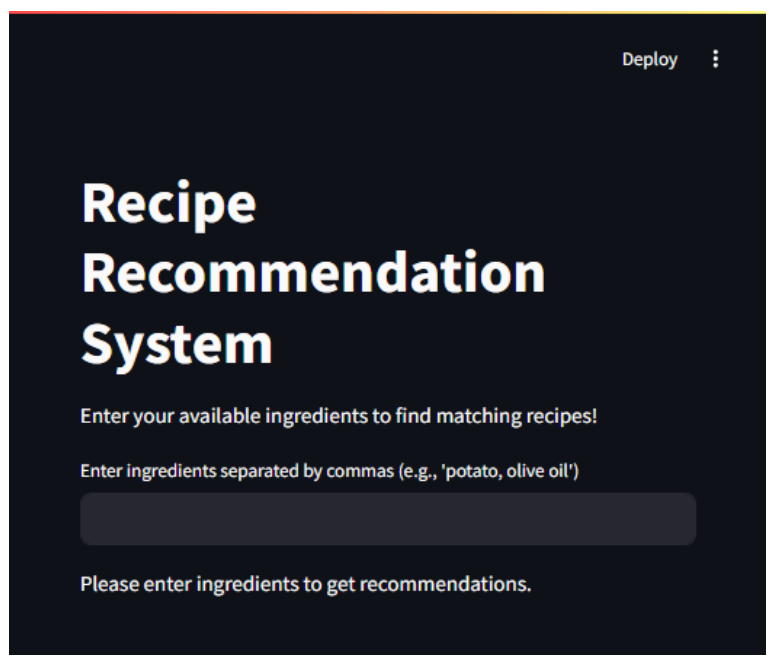

Fig. 2. Streamlit UI Dashboard

C. Front end Visualization with Streamlit:

Streamlit interface was developed to make the system interactive and user-friendly.

- With this user will be able to input the ingredients and view the recommended recipes along with instruction steps.
- It offers a seamless user experience because of it's simplicity and interactivity.

VIII. Data Analytics

The Overall effectiveness of this system in terms of user satisfaction can be evaluated based on user ratings on a scale of 1 to 10. Below are the significant findings according to Figs 3,4 and 5.

	Query	Recipe name	Response	similarity score	Rating from to 1 to 10
2	carrots, onions, bell peppers, lettuce, broccoli	After School Muscle Builder Wraps	flour tortilla,deli ham,deli turkey,onions,spinach leaves,carrots,cheese,bell peppers	0.245858017	6
3		Dehydrator All Purpose Tomato Sauce	tomatoes,green peppers,onions,carrots,garlic cloves,pimientos	0.242274036	6
4		Garden Salad (As In, the Whole Garden)	jicama,carrots,cabbage,red onions,bell peppers,romaine lettuce	0.221494561	7
5		Roast Beef With Vegetables	standing rib roast,garlic,onions,carrots,cabbage,cauliflower	0.216981019	5
6		Corned Beef and Cabbage	corned beef brisket,carrots,onions,water,cabbage	0.200480866	5
7	Potatoes, carrots, onions, garlic, parsley	Uncle Emmett's Legendary Old Original Yankee Pot Roast	chuck,salt pork,clear beef broth,parsley,thyme leaves,garlic,carrots,onions,potatoes	0.530008727	8
8		Sunday Dinner Veggies on a Baking Stone	potatoes,carrots,parsnips,onions,garlic,mushrooms,salt %26 pepper	0.448605166	7
9		Hefrid's Casserole	carrots,potatoes,onions,tomato paste,olive oil,garlic,pepper,salt,green olives,parsle	0.424853735	9
10		Abaroger Dino Curry	rice,onions,potatoes,carrots,garlic,roux,water,chicken	0.3947759085	8
11		Tasty Pork Roast	pork loin roast,onions,garlic cloves,carrots,potatoes,salt,pepper,dry onion soup mi	0.2969969091	7
12	Salmon, dill, lemon, asparagus, olive oil	Wood Smoked Salmon & Orzo Salad	orzo pasta,sea salt,fresh ground black pepper,peas,dill,lemon,flat leaf parsley,olive c	0.429848987	9
13		Grilled Asparagus	asparagus,olive oil,salt and pepper,lemon,chives	0.417002539	7
14		Baked Asparagus	asparagus,olive oil,toasted sesame seeds,salt and pepper,lemon	0.3607002342	7
15		Roasted Asparagus	asparagus,olive oil,lemon wedge	0.3603502004	6
16		SALMON WITH ARUGULA, ZUCCHINI & ASPARAGUS	salmon,olive oil,arugula,zucchini,asparagus,lemon juice,olive oil,salt %26 pepper	0.290177821	8
17	Beef, bell peppers, soy sauce, ginger, garlic	Garlic-Ginger Seitan from Scratch	vital wheat gluten flour,soy sauce,garlic,ginger	0.550969657	6
18		Marinated Grilled Flank Steak	flank steak,white wine,soy sauce,ginger,garlic	0.415343854	6
19		Russell's Quick Chicken Skewers	bamboo skewer,boneless skinless chicken breast,garlic,ginger,sake,soy sauce,chili flakes	0.329827143	6
20		Very Quick Asian Broth With Tofu and Vegetables	chicken stock,ginger,garlic,firm tofu,bok choy,cooked egg noodles,soy sauce	0.317649146	5
21		Hot! Jerky Marinade	beef round steak,soy sauce,hot sauce,gingerroot,garlic,spices	0.2938028966	4

Fig. 3. User Ratings

```
Descriptive Statistics:
count    193.000000
mean       8.243523
std        1.543728
min        4.000000
25%        7.000000
50%        9.000000
75%        9.000000
max       10.000000
Name: Ratings, dtype: float64
Mode: 9.0
Variance: 2.383095854922272
Skewness: -0.8441286828144695
Kurtosis: 0.03368476642941953
```

Fig. 4. Descriptive Statistics

A. Overall ratings trend:

- The sample mean of 8.24 signifies that the recommended recipes were received favorably by the users.
- Users were well satisfied with the recommendations as the median and mode ratings are 9.
- The rating scale has at least a rating of 4 and at most a rating of 10, with most people giving it an 8 or 9.

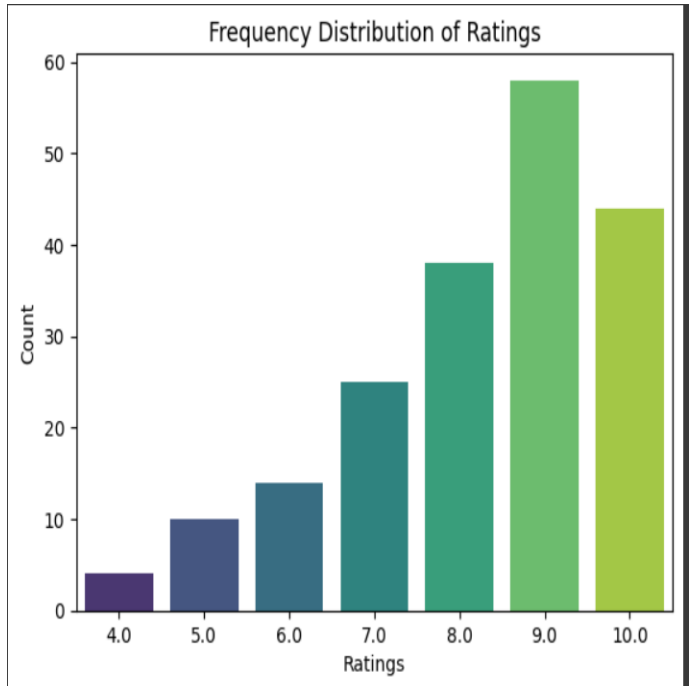

Fig. 5. Frequency Distribution

B. Variation and Distribution:

- With a standard deviation of 1.54 and a variance of 2.38. The ratings reflect a relatively wide range, with the majority dropping below 1.54 points from the average.
- A negative skew of -0.84 means that the lower end of the rating distribution can be considered small and that 9 and 10 ratings predominate in the distribution.
- It can be inferred that the distribution is normal and also does not have outliers because the kurtosis is only 0.03.

C. User Satisfaction:

As there are a higher number of ratings between the ratings 8 and 10, it is evident that the system is effective in giving satisfactory recommendations.

IX. Data Visualization and Results Report

A. Data Visualization:

In this study, user ratings for the recipe recommendation system were analyzed and shown using various visualizations. These visualizations express the system functionality and user satisfaction. We had also developed an interactive dashboard using Streamlit that would make these results easy to understand by visualizing them in real-time.

1) Histogram:: The distribution of user ratings on a scale from 1 to 10 was displayed using a histogram. As the majority of the ratings are between 8 and 10, the graph clearly skews towards high scores. This shows that most of the users were satisfied with the system.

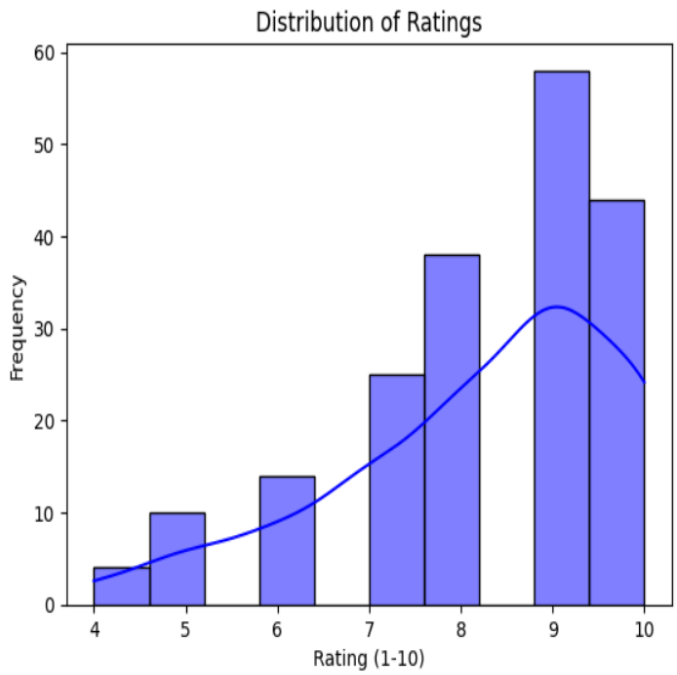

Fig. 6. Histogram

2) Box Plot:: A boxplot is used to understand how ratings are varied and spread. The interquartile range (IQR) consists of the majority of data, with a median rating of 9. As there were no outliers in the data, this proves that user feedback was consistent.

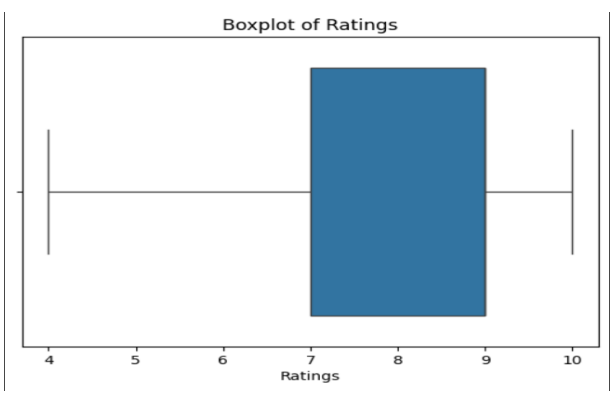

Fig. 7. Boxplot

3) Pie Chart:: Ratings were categorized into three categories: Low category contains ratings between 1 to 3, Medium category contains the ratings from 4 to 6. High category contains ratings from 7 to 10. With 85 percent of ratings falling under the category of High, this visualization further supports the system's success.

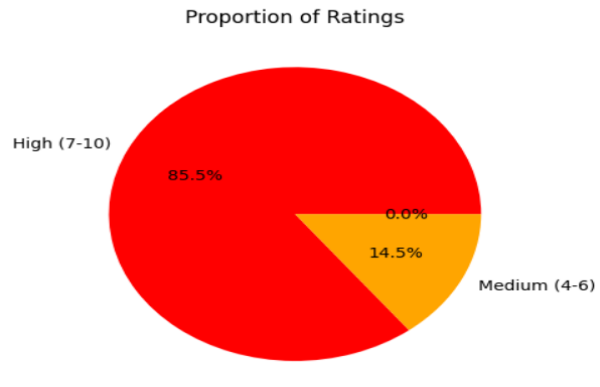

Fig. 8. Pie Chart

B. Results:

1) Streamlit dashboard outputs:: Using Streamlit, visuals were dynamically displayed on an interactive dashboard [9]. So that, users can examine the results in real-time. Apart from displaying the top k recipes, it also has a feature called "Enhance recommendations" that leverages LLM in the backend where it displays the instructions to make a recipe, suggests possible substitutions, and also why the system has recommended that particular recipe.

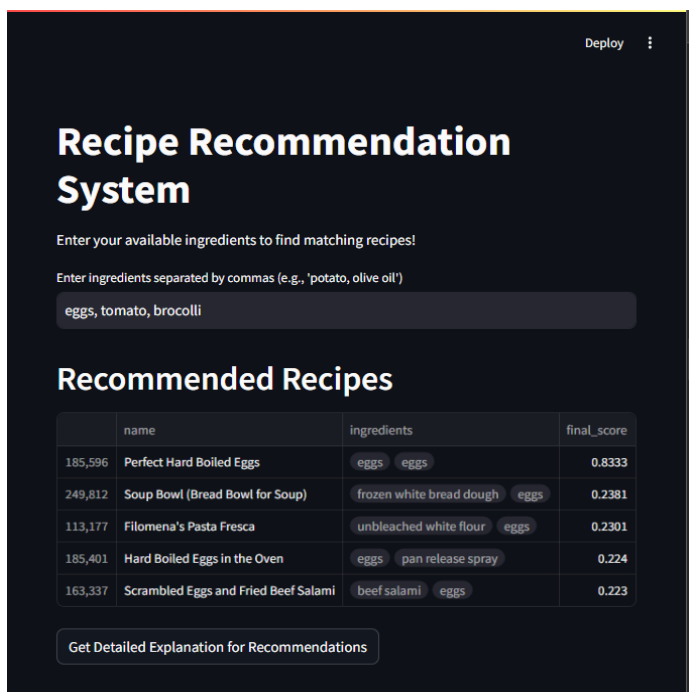

Fig. 9. Retrieval Results

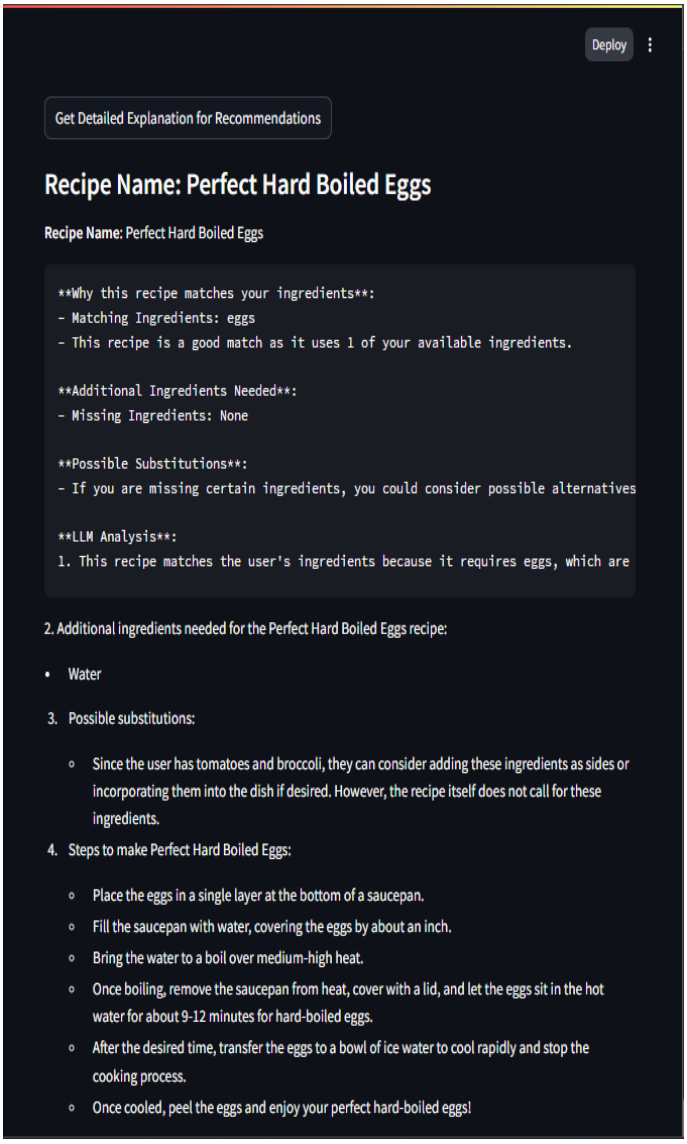

Fig. 10. LLM Results

X. Future Work

The future work for the intelligent recipe recommendation system can really have a wide range of enhancements to improve the functionality, user engagement, and adaptability. The real-time feedback that is collected can also be used to refine the recommendations i.e. the higher rated recommendations are given more importance than the low-rated ones. Incorporating nutritional and health information of the food such as calorie goals, allergies and healthy substitutions for some ingredients can be used to provide tailored health-conscious meal recommendations for the respective individuals. Also providing the users with recommendations and meal plan suggestions from practicing dieticians via a service now platform can also be included in our system.

Extending to multimodal data like images and videos can make the recommendations more engaging and visually appealing [6] while searching for the recipes based on the ingredients available so if the video is available on how to cook a particular recipe then it will inevitably be easy for the users if he/she can prepare the dish in a step by step process at their own pace and the picture can be used to identify the ingredients for the users who are in new location or doesn't know what a particular item is called. One of the technical advancements that is very interesting is to integrate the recommendation system with smart kitchen appliances and IOT data can provide real time cooking guidance and get specific recommendations based on the contents in the refrigerator. Another such development can be combining the meal planning and the shopping as this can streamline the weekly schedules and ingredients purchased for that week in the groceries application and the recommendation system specifically gives recommendations for meals to the users without the user wasting any time to decide on what to cook. The recommendations system can be also shaped to help the user to eat healthy by incorporating small milestones or challenges for the users like losing weight or walking certain number of steps this could help the user to eat healthy and also recommend the user with healthy food recommendations. All the above-mentioned improvements can help the system be one of a kind in the personalized culinary AI domain.

XI. Conclusion

The recipe recommendation system utilizes Retrieval Augmented Generation model represents a significant improvement in personalized culinary solutions, addressing the limitations of traditional systems through the latest AI techniques. By combining the enhanced data analytics, NLP and Machine Learning approaches, the system delivers a tailored recipe recommendation that caters to the specific needs of the user, by taking the inputs from the user such as ingredient availability. The usage of the extensive dataset helped to build a strong foundation for the model, the manual feedback collected from the users also helped us to calculate the metrics of the system. The proposed future enhancements which include advanced personalization, multimodal data integration could help increase the systems scalability, adaptability and relevance in recognizing the evolving user demands.

References

[1] Deldjoo, Y., He, Z., McAuley, J., Korikov, A., Sanner, S., Ramisa, A., ... & Milano, S. (2024, August). A review of modern recommender systems using generative models (gen-recsys). In Proceedings of the 30th ACM SIGKDD Conference on Knowledge Discovery and Data Mining (pp. 6448-6458).

[2] Maheshwari, S., & Chourey, M. (2019). Recipe recommendation system using machine learning models. International Research Journal of Engineering and Technology (IRJET), 6(9), 366-369.

[3] Wang, X., Wang, Z., Gao, X., Zhang, F., Wu, Y., Xu, Z., ... & Huang, X. J. (2024, November). Searching for best practices in retrieval-augmented generation. In Proceedings of the 2024 Conference on Empirical Methods in Natural Language Processing (pp. 17716-17736).

[4] Bai, X., Duan, L., Tang, R., Batra, G., & Agrawal, R. (2022, October). Improving text-based similar product recommendation for dynamic product advertising at yahoo. In Proceedings of the 31st ACM International Conference on Information & Knowledge Management (pp. 2883-2892).

[5] Shah, S., Chande, D., Pandya, K., Gopani, K., Gawade, A., Rathod, N., & Florence, A. (2024). Revolutionizing Culinary Experiences: AI-Driven Ingredient Recognition and Personalized Recipe Recommendation System. Library Progress International, 44(3), 14279-14290.

[6] Liu, G., Yin, H., Zhu, B., Chen, J., Ngo, C. W., & Jiang, Y. G. (2024). Retrieval Augmented Recipe Generation. arXiv preprint arXiv:2411.08715.

[7] Johnson, S. J., Murty, M. R., & Navakanth, I. (2024). A detailed review on word embedding techniques with emphasis on word2vec. Multimedia Tools and Applications, 83(13), 37979-38007.

[8] Douze, M., Guzhva, A., Deng, C., Johnson, J., Szilvasy, G., Mazaré, P. E., ... & Jégou, H. (2024). The faiss library. arXiv preprint arXiv:2401.08281.

[9] Richards, T. (2021). Getting Started with Streamlit for Data Science: Create and deploy Streamlit web applications from scratch in Python. Packt Publishing Ltd.

[10] Zhao, W. X., Zhou, K., Li, J., Tang, T., Wang, X., Hou, Y., ... & Wen, J. R. (2023). A survey of large language models. arXiv preprint arXiv:2303.18223.